

In pursuit of storage nirvana



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“As we produce and consume more data than we’ve ever done in history, paying for storage in the cloud will be the way to go – free services will be unsustainable for businesses and inconvenient for users after a point.”

Last month, just before the onset of the monsoon, I decided to do some late spring cleaning. As it turns out, I chanced upon an old shoebox full of CDs and DVDs from the early 2000s. After seeing the silver and golden Moser Baer discs with handwritten scribbles for titles, I was lost, gripped by nostalgia. For a second, hope led me to wonder – what if these discs still held data unscathed? I wasted no time, but alas, my data was gone forever – expensive recovery techniques aside. With that revelation, my reminiscence too disappeared, and I was left with an overwhelming sense of gloom 1.0.

I’m sure that we’ve all gone through this scenario while trying to hang onto our digital relics.

The problem is that it’s not just CDs and DVDs but even hard drives that fail after a point. When I lived in Bangalore, for instance, I purchased a 500GB external drive. Back then in 2008, I used it almost daily for over three years till I left that city in 2011. The drive sat neglected and forgotten in a cupboard of my home here in Mumbai for many months, until I plugged it in in 2012 to copy some data. The drive died on me after about a week’s sluggish performance. History repeated itself, gloom 2.0 set in.

Optical storage is limited, so is hard drive or flash storage. They all fail after a point, sooner than you think. True offline data storage isn’t a viable option, because what’s popular today will be out of fashion tomorrow. Porting data from a legacy medium to popular alternatives is ridiculously expensive today, and it will remain astronomically expensive in the future as well.

Our only immediate hope to store all our data is to spread it across cloud systems. This includes using Dropbox-like solutions, uploading photos onto Facebook, videos on YouTube, and distributed machines connected through P2P networks. As we produce and consume more data than we’ve ever done in history, paying for storage in the cloud will be the way to go – free services will be unsustainable for businesses and inconvenient for users after a point.

Let bygones be bygones. What could I do to ensure all data from this point on is preserved seamlessly, efficiently, and without pinching my pocket? How do you ensure that offline data backed up in hard drives won’t need you to pay an arm and a leg to translate into the popular medium at that point in the future? Meanwhile, what happens when all your favourite social networks or cloud storage providers start to go out of business? How will you migrate your terabytes upon terabytes of data from one source to another?

Despite storage prices falling to record lows, everything ultimately has a shelf life. And that means your data in the cloud can also “vanish” as long as that data sits with businesses that need to make a profit. Cloud storage providers are still using conventional spinning hard drives and are slowly starting to embrace SSDs. Despite announcements of Helium drives and Heat Assisted Magnetic Recording, we won’t see them for many more years. In the meantime, data is churned out left, right and centre, and there’s no way to capture it for posterity – no way to archive and freeze it forever.

M-Disc enabled optical media can apparently last a 1000 years – a claim that only future humans can verify, if they manage to find the disc in our fossil remains. But nothing else even comes close, even as far as tall marketing claims are concerned. For now, all I want is a way to be sure that whatever photos or videos I shoot of my daughter growing up will be easily visible in the next few decades off a storage medium I choose to invest in today, without too much trouble. I don’t think I will be able to handle gloom 3.0. 

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